

Reflection on Statistical Literacy and Decision-Making

1. Introduction

In a world overflowing with data and analysis, and now further amplified by Artificial Intelligence (AI), exercising critical judgment is essential for daily decision-making. Can a statistically significant conclusion have no real value, or can a confident-sounding AI output be trusted? This is where statistical literacy becomes crucial. As defined by Rumsey (2002, p.1), it is the ability to "explain, decide, judge, evaluate, and make decisions about information". It encourages us to question what lies behind the numbers, how they were sourced and analysed, and to scrutinise conclusions before trusting outcomes at face value.

2. From Concepts to Evaluations

Throughout the Numerical Analysis module, I gained valuable knowledge and practical skills in probability calculations and statistical concepts essential for descriptive and inferential analysis, as well as predictive modelling. I learned the importance of data management, assessing data quality and characteristics, and performing data transformations for further analysis. I created summary and contingency tables and developed my data plotting skills. By evaluating assumptions, I formulated and tested hypotheses using both parametric and non-parametric paths. I interpreted p-values and understood the importance of confidence intervals and sample sizes in decision-making. Additionally, I assessed relationships among variables, built linear and logistic regression models, compared model choices, and evaluated their classification performance. I concluded the module by embracing a Bayesian approach to inference—calculating posterior probabilities from prior probabilities and the likelihood—contrasting with the classic frequentist,

likelihood-based approach.

I practised and developed my understanding of statistical analysis using R, applying it to various datasets—including COVID-19 and Health Data—and capturing key insights from these formative activities in my GitHub e-portfolio (see Appendix).

3. Statistical Reasoning and Analytical Judgement

As Schober, Bossers, and Schwarte (2018) emphasise, combining descriptive and inferential statistics is essential for drawing meaningful conclusions. Ranganathan (2021) highlights that understanding the research question first is paramount. Next, she recommends analysing the data's characteristics and verifying statistical requirements—using formal tests and visual inspection—both of which are critical for valid inference. I learned that formulating a hypothesis depends on aligning with the data's underlying assumptions and understanding the type of data, the number of groups to be compared, and the relationships among them. This process guides the choice between parametric tests for comparing group means and non-parametric tests assessing the rank sums or rank differences.

However, these decisions are rarely linear. Different paths may be valid provided that assumptions are evaluated, test limitations are acknowledged, and the chosen method is well justified. For example, failing a normality test does not automatically rule out the use of a parametric test. In my blood pressure analysis, both the Mann-Whitney U test and the Welch t-test yielded the same statistical conclusion. As Fagerland (2012) notes, selecting the appropriate test is not a single-criterion decision. It requires visual inspection, assessment of outliers, and consideration of sample size and the test's robustness. Cross-validation strengthens confidence in these decisions.

Furthermore, these findings—supported by McShane *et al.* (2024)—confirm that statistical significance is not a simple, binary decision. Greenland *et al.* (2016) and Tijssen (2021) argue that the p-value alone does not convey meaningful information. While it provides evidence against the null hypothesis, it does not address the magnitude, precision, or practical significance of an observed effect, which are captured by the confidence interval.

My R activities confirmed the existence of this gap. In my multilinear regression model, one variable had a borderline p-value near the 0.05 significance level. The wide confidence interval showed the effect was real, but the effect size was imprecise. Conversely, a demographic predictor was significant in a logistic model but became insignificant after adding another variable. The odds ratio and a narrow confidence interval indicated that the effect was precise but negligible. These examples demonstrate why considering the full picture reveals insights, strengthens conclusions and avoids misinterpretations.

Additionally, summary and contingency tables form the foundation of analysis. While extracting summary output in R is straightforward, presenting it meaningfully requires careful consideration. Choosing between means/standard deviations and medians/interquartile ranges for continuous variables, and between counts, percentages, or both for categorical variables, should always align with the purpose of analysis.

Contingency tables, meanwhile, prompt important questions about how categorical variables are allocated. I came to understand that their analytical value and interpretation depend on their organisation and layout, which should be guided by the research question rather than by R's default output.

Finally, I realised that creating graphical visualisations is more than just producing graphs. My experience and peer feedback—such as from Brown's forum discussion—taught me that careful data preparation, thoughtful selection, and simplicity are key. The abundance of graphic choices challenged my restraint, and I learned the importance of aligning visuals to the audience's graphic literacy.

My perspective shifted from viewing graphs as mere outputs to appreciating them as intelligent analytical reasoning tools. As emphasised by Balan (2024), when used effectively, visualisation promotes understanding, powerfully conveys a clear message, builds trust with the reader, and aids decision-making.

4. Moving Forward

Given my professional experience as a Product Manager collaborating with data analysts to drive business outcomes, I am committed to refining my analytical approach. Equipped with statistical tools and frameworks, I can serve as a valuable sparring partner, constructively and confidently challenging analysts' outputs. Instead of accepting pre-produced analysis at face value, I can assess the statistical validity of results, look beyond “significant/no-significant” statements, probe for assumption checks, confidence intervals, and the representativeness of sample sizes, and recognise when a deeper dive is needed. Furthermore, I can critically evaluate findings based solely on correlation and investigate whether other correlated factors may underlie the apparent results, helping to avoid costly business decisions.

Moreover, I came to the realisation that statistical terms, such as “significant”, “proxy”, and “correlation vs causation”, are often used without proper understanding. This awareness enables me to share and promote statistical knowledge within the product community. I can confidently support decisions grounded in robust evidence rather

than surface-level significance, which is especially important in our AI-saturated era, where critical judgment is more vital than ever.

5. Conclusion

Taking the Statistical Analysis course has transformed my relationship with statistics. Instead of simply trusting statistical methods, I now feel empowered to question and scrutinise them. While R offers mathematically sound answers, it is essential to challenge assumptions, understand test limitations, and carefully interpret statistical results within the research context and their practical relevance. This course was not about becoming a statistician, but about developing the ability to apply statistical judgment and to ask better questions that contribute to reliable, trustworthy, and effective decisions grounded in statistical evidence.

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Appendix

GitHub e-portfolio: <https://agadjg.github.io/eportfolio/pages/numerical-analysis.html>